

CSI-Sense: Detecting Presence Behind Obstructions For S.A.R. Using WiFi



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BACKGROUND Pt 1.

How does Wifi work?

Sends bouncing radio waves around the house from the transmitter (router)

Signals can change from slightest of movement

RSSI	vs.	CSI
Single Dimensional		Channel State Info Multi Dimensional
Amount of power when received by receiver		Amplitude and Phase of each tiny frequency
		Amplitude → Measurement of strength Phase → Measurement of “timing”

Other important features of Wifi

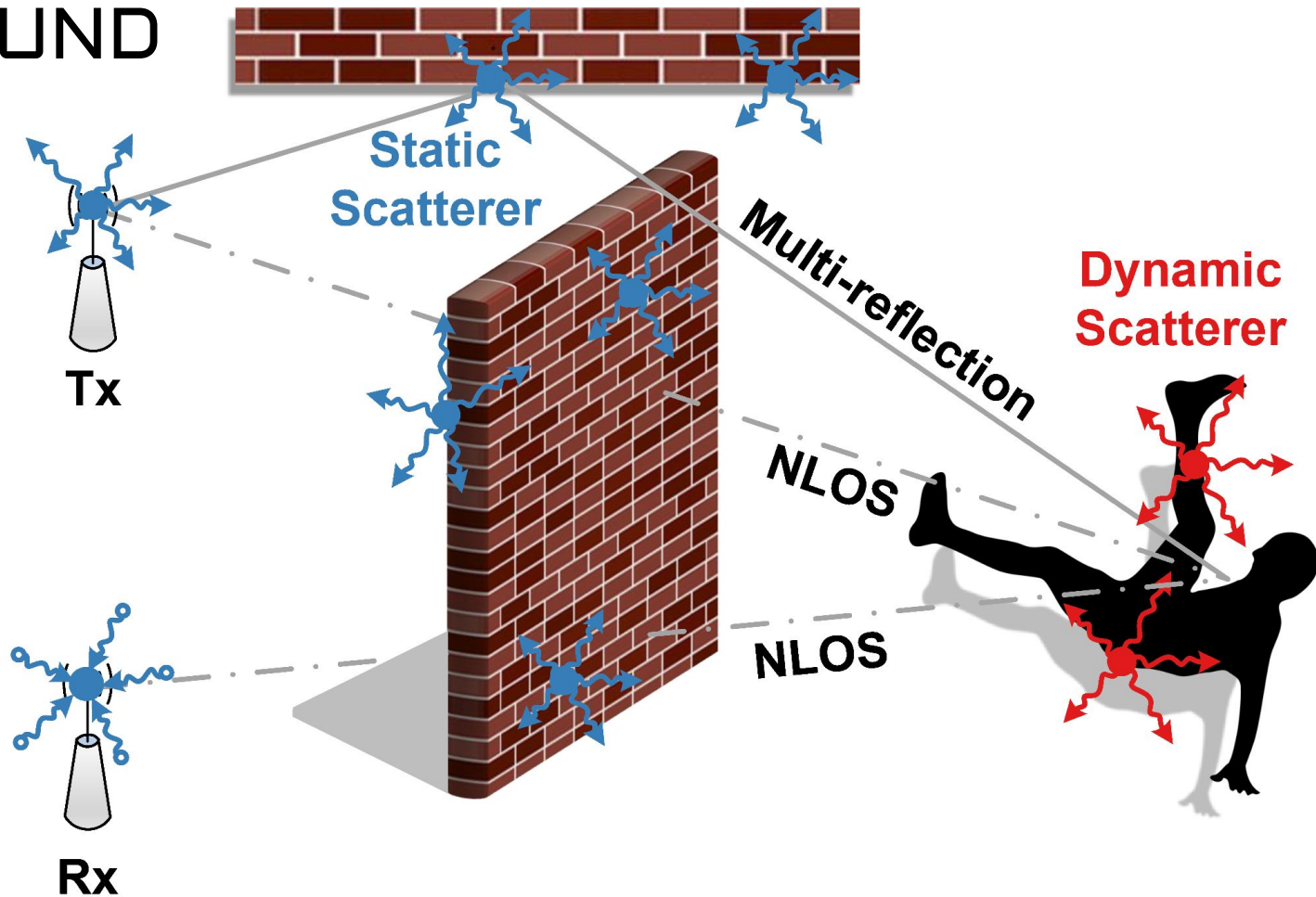
- 2.4 Ghz penetrates walls the best
- Uses Orthogonal Frequency-Division Multiplexing (OFDM)
 - Splits signal into hundreds of subcarriers
- Can penetrate dozens of meters, depends on amount of obstruction

Depending on frequency, can pass through walls!



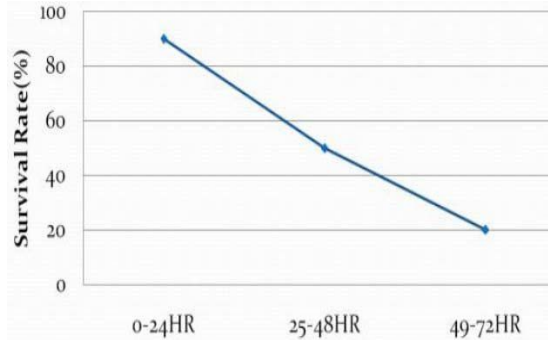
BACKGROUND

Pt 2.



Problem - Structural Search & Rescue

- Collapsed structures
- Hazardous & limited access for rescuers
- Time Critical: Survival Rates drop after the first 24-72 Hours
- Obstruction Visibility: Darkness, smoke, dust, debris
- Victims may be unconscious therefore unable to signal
- Need reliable detection **without line of sight**



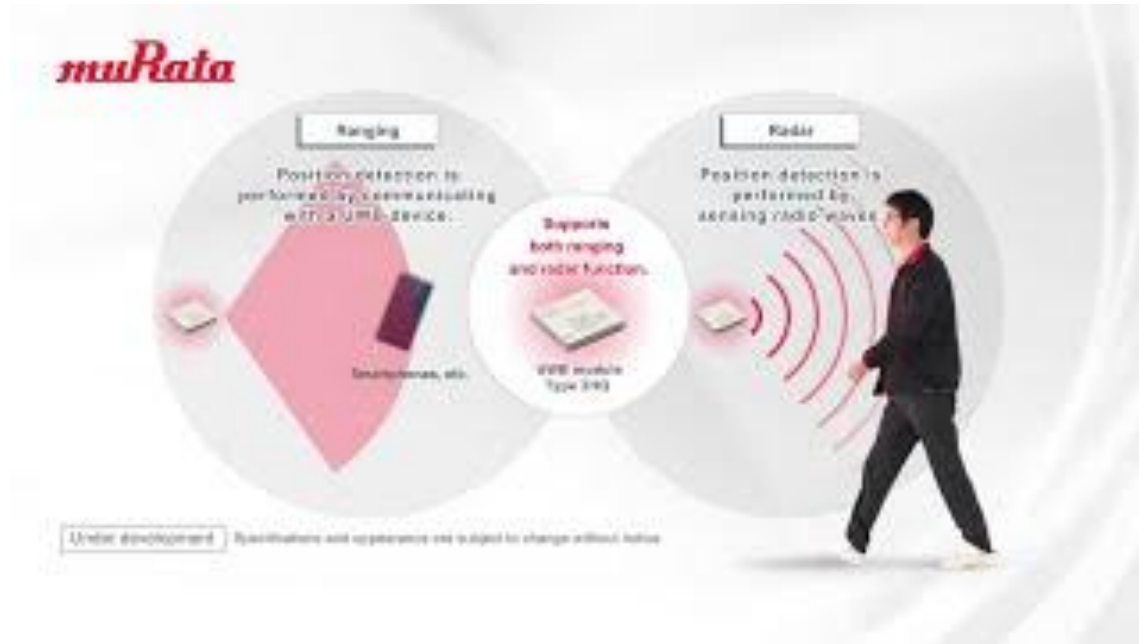
OTHER SOLUTIONS

- Thermal cameras: detect heat but cannot see through solid walls or heavy debris & are affected by environment
- Acoustic Sonar Devices: mainly underwater rescues
- Rescue dogs: effective but slow, resource-intensive, and operationally constrained
- Commercial UWB/through-wall radar: Tens of thousands of dollars



MULTI-APP BOX

Quick ~20 Second Demo of UWB



Novelty

I am developing a Wifi based SAR system to detect presence through rubble using CSI data

Initial setup is using a Raspberry Pi 4B and a router to test detection through different obstacles

Uses existing WiFi hardware → no need for specialized, expensive sensors like UWB radar or thermal cameras.

Portable and easily deployable → multiple units can be set up quickly during disaster response

Real-time monitoring and integration into current search and rescue workflows



Impact

- A tool like this will hopefully be able to be used by small scale responders
 - Things like NGOs (non-governmental organizations), firefighters, local emergency teams
- Improved responder safety → participating in SAR can be a dangerous task, being able to scan from distance reduces risk
- More scalable Search & Rescue Operations, better for large structural damage



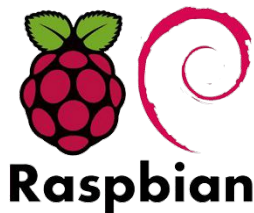
Why is This Better?

- Low-cost setup (~\$150–200) using widely available components like Raspberry Pi and Wifi router



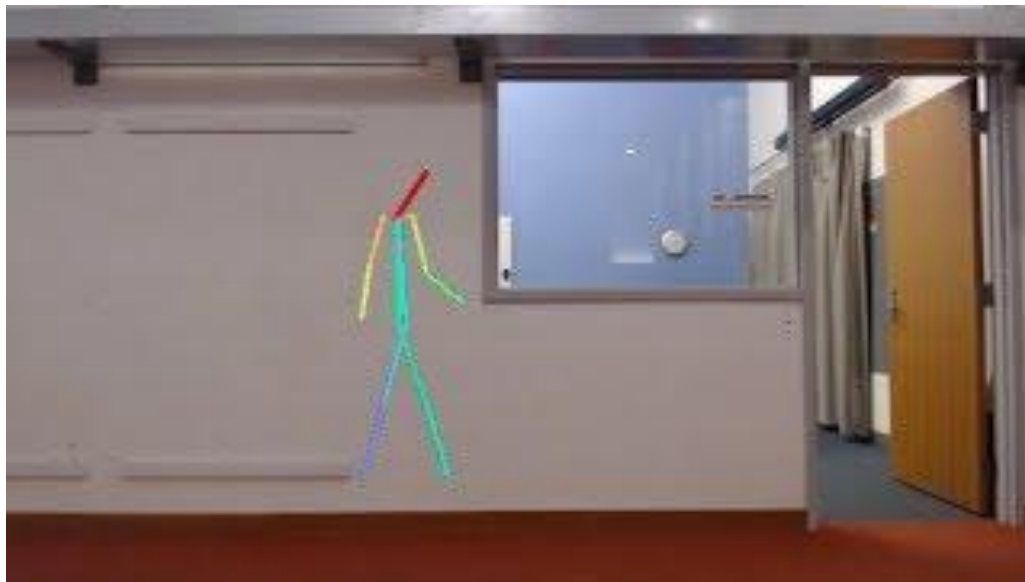
- No specialized radar or bulky commercial hardware required
- Unlike some studies, I'm completely building upon open source technology so everything is adaptable for future works

(UWB radar)

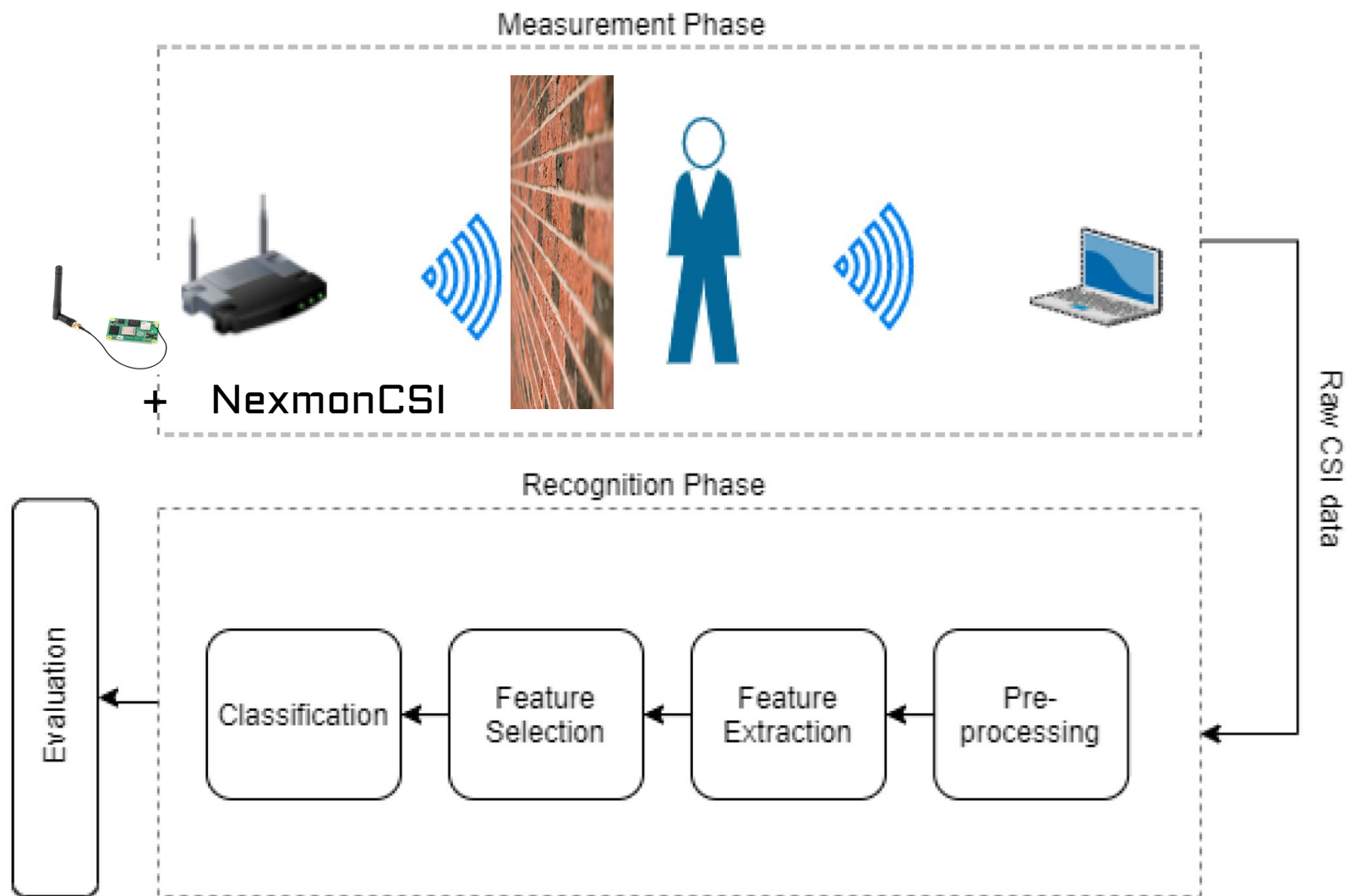


nexmon

MIT Demo: RF-Pose

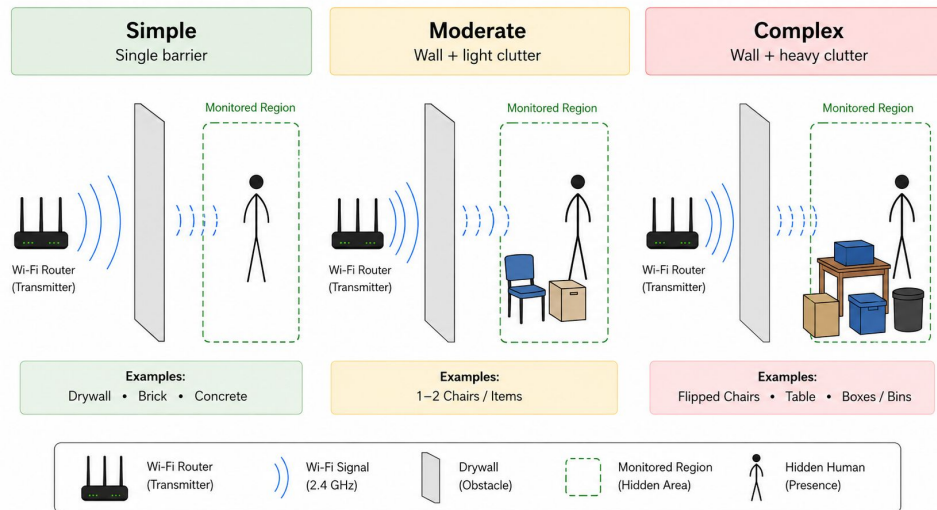


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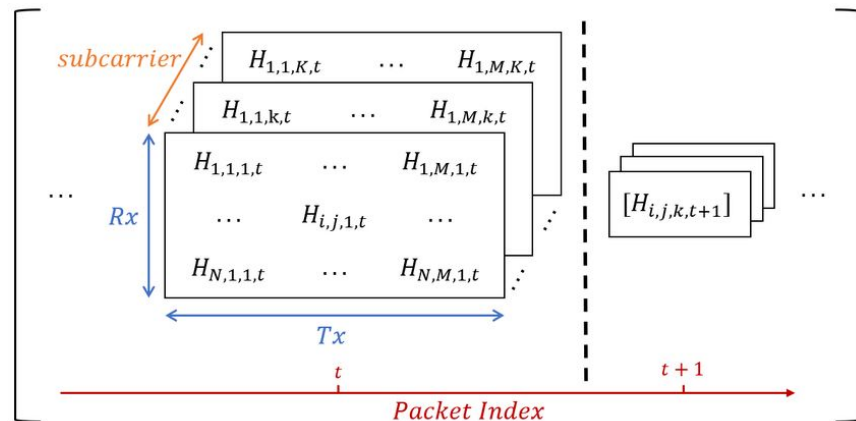
Measurement Phase

- Monitored region will have different complexities
 - Simple: singular wall; drywall, brick
 - Moderate: Wall + light clutter → 1-2 chairs/items
 - Complex: Wall + heavy clutter → flipped chairs, table, boxes/bins
- Baseline vs human for each obstruction
- Save these CSI logs (e.g., PCAP files) to a laptop for later preprocessing

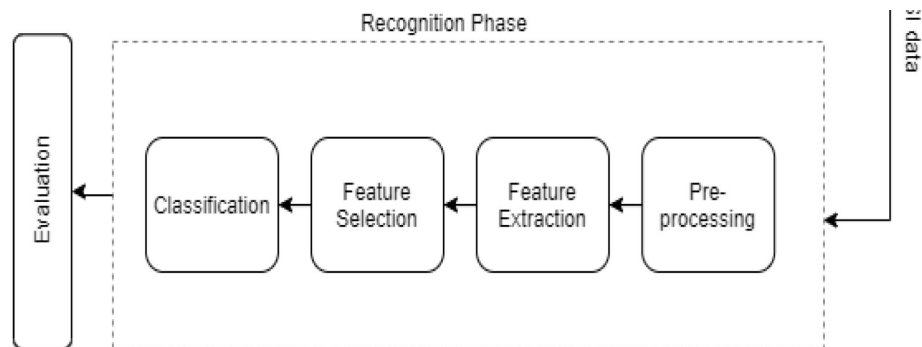


Recognition Phase

- Pre-processing
 - Take the PCAP files from the Pi and convert them into CSI matrices
 - Throw out obviously corrupted subcarriers → will show in demo
- Time-Series Segmentation
 - Chop each CSI recording into fixed-length windows (1.5-second segment of CSI data); each window is one instance
 - What features: Will explain in demo 2
- Classification
 - Random Forest to do binary classification: 0 = baseline, 1 = human presence.



- $H = a+jb \rightarrow$ one "CSI" scalar
- $|H| = \text{sqrt}(a^2+b^2) \rightarrow$ amplitude
- $\theta = \text{atan2}(b, a) \rightarrow$ phase



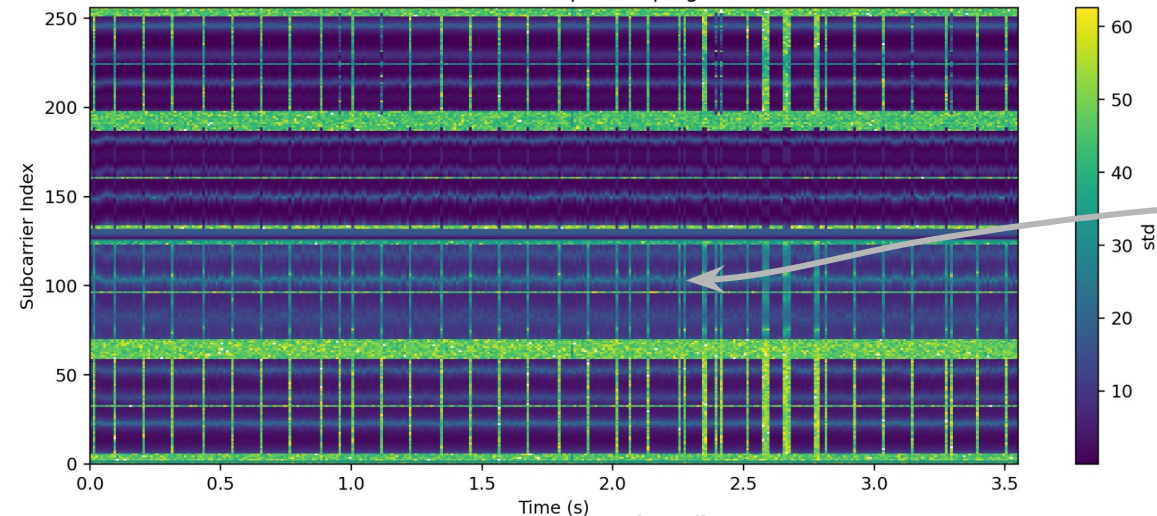
Pipeline Demo 1

Test data: Glenn Forbes. (2021). Dataset: WiFi-based Human Activity Recognition using

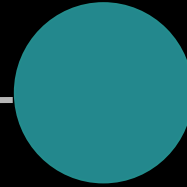
Raspberry Pi (1.0) [Data set]. Zenodo.

Labeled the files based on different human actions

Forbes Heatmap - Sleeping

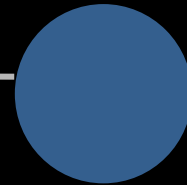
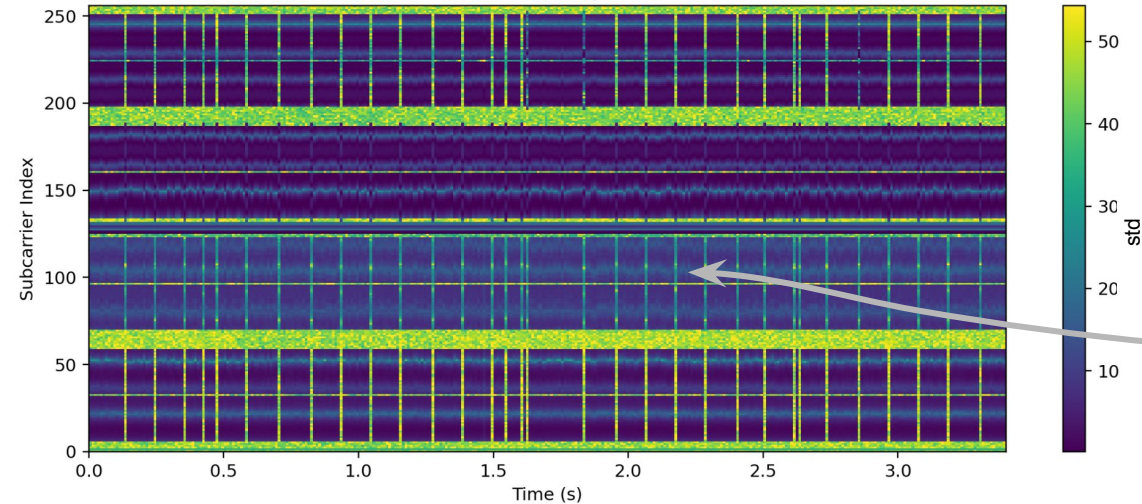


Std - standard deviation is measuring how much CSI is fluctuating!



- Around subcarrier 110 at 2.35 s, sleeping has much larger fluctuations than the empty baseline.
- sleeping is about 31, baseline is about 17, meaning a quiet human (just breathing) nearly doubles the variability on that subcarrier.

Forbes Heatmap - baseline



Model Information

Analogy: playing a chord on piano has low, mid, high notes that exist simultaneously → dividing up those notes

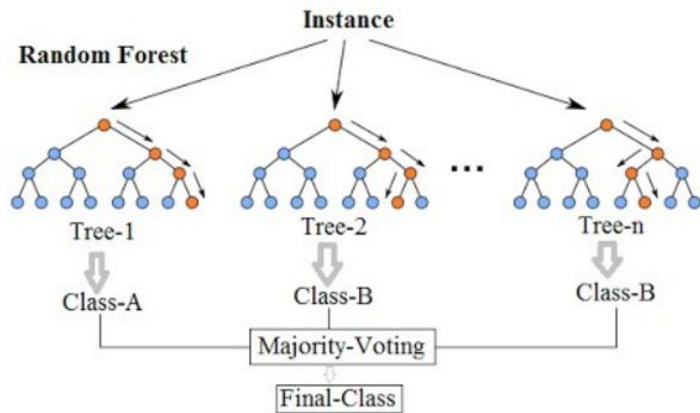
- **I/O**

- Pcap files, CSI matrix
- A heatmap video, red blob with changing brightness based on intensity

- **Model**

- Binary Random Forest Classifier

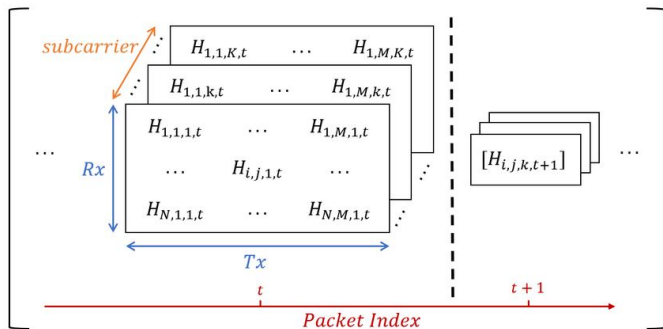
Random Forest Simplified



- **Features and Model**

- Low-frequency energy fraction (low)
 - Power in frequencies $f \leq 0.5$ Hz divided by total power in the window. Captures very slow changes (almost static channel).
- Mid-frequency energy fraction (mid)
 - Power in frequencies $0.5 < f \leq 2.5$ Hz divided by total power. Region where breathing and slow body sway tend to show up.
- High-frequency energy fraction (high)
 - Power in frequencies $f > 2.5$ Hz (up to $\text{MAX_FREQ} = 10$ Hz) divided by total power. Captures faster motion like walking or arm movements.
- Spectral entropy (ent)
 - Entropy of the power distribution across frequency bins.
 - Low entropy = power concentrated in a few frequencies; high entropy = power spread out (messier motion pattern).

Explanation



NexmonCSI already collapses the Rx and Tx, so instead of $H_{i,j,k,t}$ its $H_{k,t}$

motion directly affects **both**:

- *how fast* the signal varies
- *how much* it varies

Low power + low Hz = low motion

Vice versa

Step 1. Collapsing Subcarriers

```
amp = np.abs(csi)
```

```
sig[t] = mean over subcarriers ← how the channel strength changes over time
```

Step 2. Creating Temporal Frequency (Hz)

How often does the signal oscillate per second?

Using short-time Fourier transform (STFT)

$$X(f) = \sum x[t]e^{-j2\pi ft}$$

Step 3. Finding power (how much signal varies)

Power is the average squared magnitude of the rate of change

$$\text{Power} = \frac{1}{N} \sum_t |v[t]|^2$$

Therefore smaller Hz → less motion detected

Step 4. Normalize power

Divide the raw power by the total power to maintain ratios

Ex. if someone breaths from far away vs. close, close will seem to have higher amplitude even though it's the same actions

Allows model to not confuse distance

===== TEST FILE-LEVEL ACCURACY REPORT =====

Overall: 90/100 = 0.900

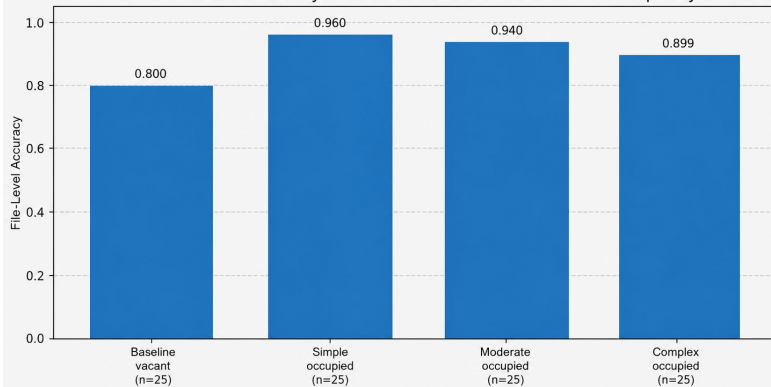
VACANT : 20/25 = 0.800

OCCUPIED: 70/75 = 0.933

Classification report (file-level, test split):

	precision	recall	f1-score	support
vacant	0.80	0.80	0.80	25
occupied	0.93	0.93	0.93	75
accuracy			0.90	100
macro avg	0.87	0.87	0.87	100
weighted avg	0.90	0.90	0.90	100

CSI-Sense Accuracy: Baseline vs Estimated Obstruction Complexity



Baseline uses vacant files. Complexity groups use occupied files only and are estimated from CSI signal distortion.



File: DD4.pkl

Actual: VACANT

Pred: VACANT (0.03)

Runtime: 0.0s

Limitations

- Cannot replicate true SAR conditions:
 - Testing environment lacks the chaos of actual disaster scenarios
- Limited to simple wall/obstacle configurations rather than multi-layered rubble
- Larger, more comprehensive data set
 - More locations should be tested instead of just my house

Future Work

- Systematic study of RF penetration through:
 - different construction materials
 - Material thickness & distance
- Direct detection rather than pi→laptop
- A better display system rather than side by side video.

REFERENCES

- 1.) <https://people.csail.mit.edu/fadel/wivi/evaluation2.html>
- 2.) <https://www.sciencedirect.com/science/article/pii/S0952197623013556>
- 3.) <https://arxiv.org/abs/2301.00250>
- 4.) https://openaccess.thecvf.com/content_cvpr_2018/papers/Zhao_Through-Wall_Human_Pose_CVPR_2018_paper.pdf
- 5.) <https://github.com/saidElhadi/esp32-mesh-csi/tree/main/monitor>
- 6.) https://www.researchgate.net/publication/356293965_Motion_Detection_using_CSI_from_Raspberry_Pi_4
- 7.) https://github.com/seemoo-lab/nexmon_csi
- 8.) https://www.researchgate.net/figure/Statistics-of-Rescue-Time-and-Survival-Rate-5_fig2_259674878
- 9.) [Glenn Forbes. \(2021\). Dataset: WiFi-based Human Activity Recognition using Raspberry Pi \(1.0\) \[Data set\]. Zenodo.](#)
- 10.) <https://dl.acm.org/doi/10.1145/2746285.2755969>

THANKS!
QUESTIONS?